

Gurudatta Sai Prasad Akkala

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 6 years expertise in architecting AWS and Azure data pipelines, slashing processing times by 65% and shrinking manual effort by 70%. Delivered real-time fraud-detection and compliance solutions, transforming terabytes of streaming and batch data into actionable BI insights. Skilled at leading cross-functional teams from prototype through production, driving cost savings and accelerating time-to-value.

SKILLS

- **Programming Languages:** C, C++, Python, SQL, R, Java, JavaScript, HTML
- **Python Libraries:** NumPy, Pandas, NLTK, Matplotlib, Seaborn
- **Frameworks:** Flask, Django
- **Cloud Platforms:** Amazon Web Services (Redshift, Lambda, Glue, S3, EC2), Google Cloud Platform (GCP), Microsoft Azure
- **Big Data Technologies:** PySpark, Apache Spark, Hadoop, Hive, HDFS, HBase, Kafka, Databricks, MapReduce, Scala, Pig
- **ETL Tools:** Airflow, Apache NiFi, Sqoop, Flume, Oozie
- **Databases:** SQL Server, PostgreSQL, MySQL, Oracle, Snowflake, MongoDB, DynamoDB, Amazon Redshift, Workbench
- **Visualization Tools:** Tableau, Power BI
- **Security Tools:** Snort, Nmap, Nessus, Burp Suite, Metasploit, Wireshark
- **DevOps & CI/CD:** Jenkins, Docker, Kubernetes, Git, GitHub, Terraform, Gradle, Maven
- **Applications & Tools:** Amazon Connect, Data Warehousing
- **IDEs & Operating Systems:** Eclipse, IntelliJ, PowerShell, Visual Studio, Windows, Linux, UNIX
- **Methodologies:** Agile/Scrum, Waterfall

WORK EXPERIENCE

Comcast | Data Engineer | Philadelphia

July 2024 – Present

- Streamlined real-time fraud detection pipelines by optimizing data ingestion and preprocessing workflows, reducing end-to-end latency and improving anomaly detection accuracy across high-velocity telecom and entertainment data streams.
- Deployed scalable Apache Spark and Airflow pipelines to process terabytes of daily call detail records (CDRs) from AWS S3, enabling real-time threat detection and customer verification; enforced data governance with PII masking and role-based access control.
- Enhanced hybrid batch and streaming ETL workflows using Airflow DAGs, integrating Python for processing and Kafka for real-time event ingestion; automated detection of subscription fraud and SIM swap attacks, reducing manual intervention by 70%.
- Optimized Snowflake performance through time-based and customer ID partitioning, improving query response times; enhanced Matillion ETL pipelines with parallel and incremental loads to increase throughput and reduce compute cost.

Cross River | Associate Data Engineer Intern | New Jersey

January 2023 - August 2023

- Improved a real-time fraud detection pipeline in Snowflake by streamlining data ingestion and preprocessing, reducing execution time by 40% and increasing accuracy by 25% in detecting suspicious transactions and account anomalies across banking systems.
- Managed Spark-Airflow pipelines to process 1TB+ of transactional data daily, ensuring 95% SLA reliability for real-time AML monitoring of high-risk patterns. Enhanced customer segmentation for KYC initiatives and supported compliance with BSA and FATF standards.
- Enhanced ETL pipelines using Airflow, Python (parallel processing, incremental reads), and Kafka to automate batch and real-time data ingestion into Snowflake, cutting manual effort by 70% and improving reporting latency by 3 hours for critical banking analytics.
- Augmented Snowflake (v6.0) schemas with clustering on timestamp and customer_id, enabling partition pruning and reducing query latency by 65%. Streamlined Matillion ETL (v2.44) by implementing parallel loading and incremental CDC logic.

HealthPlix | Data Engineer | Bengaluru

July 2020 – July 2022

- Developed a scalable AWS data lake for 100TB+ of healthcare insurance data using EMR (Spark), S3, and Redshift; integrated AWS Glue Catalog to ensure GLBA compliance and cut analytics onboarding time by 30%.
- Fine-tuned Spark workloads on AWS EMR, reducing data transformation time by 50% and cloud costs by 70%. Optimized Kafka-based Change Data Capture (CDC) pipelines to ingest real-time healthcare policy and claims updates with minimal latency.
- Executed Spark-based data pipelines to process healthcare claims and enrollment data, supporting ML models (Random Forest, XGBoost) to predict member preferences. Enabled a 15% boost in engagement and 10% churn reduction across key care programs.
- Implemented a HIPAA-compliant serverless AWS data framework (Lambda, IAM roles, DynamoDB with KMS encryption, Kerberos); automated access-control audits and enforced encryption at rest/in transit, reducing unauthorized-access risk by 80%.

Digit Insurance | Data Analyst | Hyderabad

July 2018 – June 2020

- Collaborated with underwriters, IT and compliance to revamp the Policy Management System—cutting manual rework by 30%, speeding up policy issuance cycles by 20%, and delivering real-time dashboards for underwriting decisions.
- Partnered with data science and claims teams to build a dynamic scoring engine in T-SQL and Python, flagging the top 5% of high-risk claims. Increased monitoring efficiency by 40% and cut average investigation time from 72 to under 6 hours.
- Refined ER/Win schemas to integrate policy, claims, and customer data—boosting consistency by 25%, eliminating redundant tables, and standardizing definitions across underwriting and actuarial systems for faster analytics.
- Streamlined complex T-SQL stored procedures and SSIS workflows to ingest, cleanse and validate claims data—cutting end-to-end processing time by 30% and accelerating monthly report delivery by 35% for operations and exec stakeholders.

EDUCATION

Master of Science in Management Information Systems

Stevens Institute of Technology, School of Business, Hoboken, NJ

May 2024

Master of Computer Applications

Acharya Nagarjuna University, Guntur, India

June 2022

Bachelor of Computer Science

Acharya Nagarjuna University, Guntur, India

March 2020

PROJECTS

Network Intrusion Detection System (NIDS)

September 2023 – December 2023

- Developed a PySpark ETL on a 10-node Hadoop cluster to process 5 TB of UNSW-NB15 traffic into Parquet.
- Implemented Random Forest & SVM models for 92% threat detection, deploying via Docker & Airflow to raise reliability by 20%.
- Optimized Spark executors and HDFS settings, cutting run-times by 25% and lifting throughput by 30%.

Vulnerability Assessment & Penetration Testing (VAPT)

January 2023 – April 2023

- Developed an automated vulnerability scanning workflow using AWS Lambda, integrating Burp Suite and OWASP ZAP to schedule
- Orchestrated periodic security scans, reducing manual effort by 60% and enabling real-time log ingestion for audit analysis.

Secure Web Application Development

September 2022 – December 2022

- Integrated 2FA and OAuth 2.0 authentication into a data-sensitive e-commerce platform.
- conducted code-level audits to detect and remediate SQL injection and XSS vulnerabilities
- Reduced security risks by 25% and ensuring data integrity across transactional pipelines.

CERTIFICATIONS

- AWS Fundamentals, Migrating to AWS Cloud, Architecting Solutions on AWS, AWS Cloud Technical Essentials, Building Data Lakes on AWS, Exam Prep: AWS Certified Solutions Architect - Associate | Amazon Web Services
- Visualize real-time data with Azure Stream Analytics and Power BI | Microsoft
- Python for Data Science, AI & Development | IBM, Crash Course on Python | Google
- Databases and SQL for Data Science with Python | IBM, SQL | Sololearn
- Hands - On Tableau Training for Data Science | Udemy
- Complete Microsoft Power BI Bootcamp | Udemy

LEADERSHIP

- Led a team in the Strategy Simulation: Competitive Dynamics and Wintel V2, achieving a 15% increase in simulated market share.
 - Fine-tuned collaboration and optimized resource allocation to outperform simulation benchmarks.

February 2022